9.3 Combining Two Functions: Products

A Definitions

The product of two functions is defined by

$$(fg)(x) = f(x)g(x)$$
$$(f \times g)(x) = f(x) \times g(x)$$

Ex 1. Let $f(x) = x^2$ and $g(x) = \log x$. Find

a) $(fg)(1) = f(1)\,g(1)$
$= 1\,(0) = 0$

b) $(f \times g)(0) = f(0)\,g(0)$
$= 0$ (undef) $=$ undefined

B Domain of the Product of Two Functions

The domain of the product of two functions is the intersection of their domains.

$$D_{fg} = D_{f \times g} = D_f \cap D_g$$

Ex 2. For each case, find the domain of the product of the given functions.

a) $f(x) = 2^x$; $g(x) = \sin x$

$D_f = \mathbb{R}$; $D_g = \mathbb{R}$
$D_{fg} = \mathbb{R}$

b) $f(x) = \dfrac{1}{x-2}$; $g(x) = \log x$

$D_f: x \neq 2$; $D_g: x > 0$
$D_{fg} = (0, 2) \cup (2, \infty)$

c) $f(x) = \sqrt{3-x}$; $g(x) = \log(x+1)$

$D_f: x \leq 3$

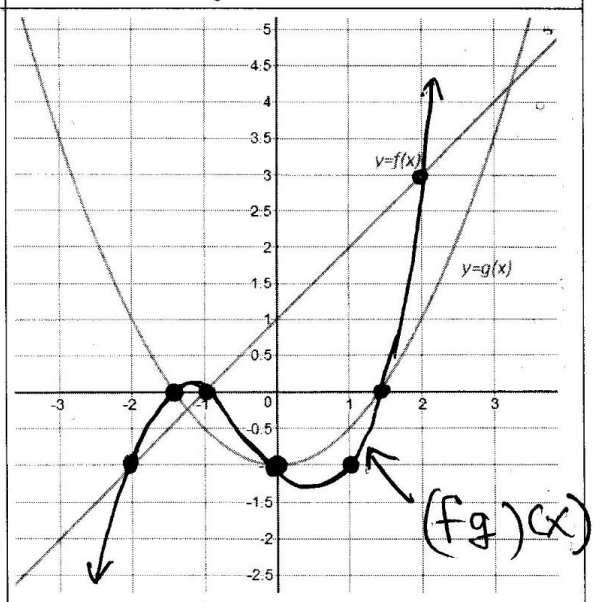
$D_g: x > -1$
$D_{fg} = (-1, 3]$

C Point by Point

Evaluate fg at every possible number x.

Note.
f is linear
g is quadratic
fg is cubic

Ex 3. The functions f and g are given by their graphs on the right figure. Graph the function fg.



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Ex 4. Prove that the product of two odd functions is an even function.

f, g are odd

$(fg)(x) = \underbrace{f(-x)}_{-f(x)}\underbrace{g(-x)}_{-g(x)}$

$= f(x)g(x) = (fg)(x)$

Ex 5. Complete the following table. Justify your reasoning.

	fg	even	odd
g	even	even	odd
	odd	odd	even

Ex 6. For each case, justify your answer.

a) Is the product of two polynomial functions a polynomial function?

yes

b) Is the product of two rational functions a rational function?

yes

c) Is the product of two sine functions a sine function? SKIP

d) Is the product of two periodic functions a periodic function?

SKIP

Ex 7. For each case, graph on the same grid the functions f and g, and then the graph of the product fg. Use technology (Desmos) to check your answer.

a) $f(x) = x$; $g(x) = \sin x$

b) $f(x) = \cos x$; $g(x) = \sqrt{x}$

c) $f(x) = x^3$; $g(x) = 0.5^x$

d) $f(x) = \log x$; $g(x) = 10^x$

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Ex 8. Find the zeroes of the product fg where

$f(x) = \sqrt{x^2 - 4x}$; $g(x) = \log(2 - x)$

$(fg)(x) = f(x)g(x) = 0 \Rightarrow$

Possible $f(x) = 0$ or $g(x) = 0$

Zeros: $\underbrace{0, 4}_{f(x)=0}, \underbrace{1}_{g(x)=0}$

$4 \notin D_g$

$1 \notin D_f$

$\therefore$ Only $x = 0$ is a zero of fg

Ex 9. If D_f and D_g are the domains of the functions f and g and if Z_f and Z_g are the sets of zeroes of the functions f and g, make a statement about the zeroes of the product fg.

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Reading: Nelson Textbook, Pages 531-537
Homework: Nelson Textbook, Page 537 #3, 5, 8, 12, 17

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